Decimal Arithmetic in $\mathbb{Q}$

1. Evaluate $14.78 + 6.954$.
2. Evaluate $23.4 - 8.67$.
3. Evaluate $-12.35 + 4.8$.
4. Evaluate 3.4×2.6.
5. Evaluate -0.45×1.8.
6. Evaluate $19.32 \div 6$.
7. Evaluate $4.68 \div 0.12$.
8. A scientist combines two liquid samples with masses of 18.45 g and 7.825 g, then removes 4.9 g for testing. What is the remaining mass of the mixture?
9. A rectangular metal plate measures 3.25 m in length and 1.4 m in width. Calculate the area of the plate in square meters.
10. A spool containing 15.75 m of copper wire is cut into equal segments of length 0.45 m each. How many segments can be cut from the spool?

Solutions

1. 21.734

   Explanation: Align decimal points, pad with trailing zeros, and add column by column:

$$\begin{array}{r} 14.780 \\ +\,6.954 \\ \hline 21.734 \end{array}$$

2. 14.73

   Explanation: Align decimal points, write 23.4 as 23.40, and subtract:

$$\begin{array}{r} 2\not{3}^{1}.\not{4}^{3\,1}0 \\ -\ 8.\ 6\ 7 \\ \hline 14.\ 7\ 3 \end{array}$$

3. -7.55

   Explanation: Subtract the smaller magnitude from the larger magnitude, retaining the negative sign:

$$\begin{array}{r} \not{1}^{0}\not{2}^{1}.\not{3}^{2\,1}5 \\ -\ 4.\ 8\ 0 \\ \hline 7.\ 5\ 5 \end{array}$$

$$\implies -12.35 + 4.8 = -7.55.$$

4. 8.84

   Explanation: Multiply both numbers by 10 to convert them into whole numbers (34×26), perform the multiplication, and then divide the result by 100 (10×10):

$$3.4 \times 2.6 = \frac{34 \times 26}{100} = \frac{884}{100} = 8.84.$$

$$\begin{array}{r} {\scriptstyle 2\ \ } \\ 34 \\ \times\,26 \\ \hline 204 \\ 680 \\ \hline 884 \end{array}$$

5. -0.81

   Explanation: Convert to whole numbers by multiplying by 100 and 10 respectively (total factor of 1000), perform the multiplication, divide by 1000, and apply the negative sign:

$$-0.45 \times 1.8 = -\left(\frac{45 \times 18}{1000}\right) = -\left(\frac{810}{1000}\right) = -0.81.$$

$$\begin{array}{r} {\scriptstyle 4} \\ 45 \\ \times 18 \\ \hline 360 \\ 450 \\ \hline {\scriptstyle 1} \\ 810 \end{array}$$

6. 3.22

   Explanation: Divide 19.32 by 6 using the bus stop method:

$$\begin{array}{r} 3.22 \\ \hline 6\overline{)19.^{1}3^{1}2} \end{array}$$

   Thus, $19.32 \div 6 = 3.22$.

7. 39

   Explanation: Scale both numbers by 100 to get $468 \div 12$, then divide:

$$\begin{array}{r} 39 \\ \hline 12\overline{)46^{10}8} \end{array}$$

   Thus, $4.68 \div 0.12 = 39$.

8. 21.375 **g**

   Explanation: Add the two initial masses, then subtract the removed sample:

$$\begin{array}{r} 18.450 \\ +\,7.825 \\ \hline {\scriptstyle 1\,1} \\ 26.275 \end{array} \qquad \begin{array}{r} 2^{1}\not{6}^{5}.2^{1}75 \\ -\ 4.\ 900 \\ \hline 21.\ 375 \end{array}$$

   The remaining mass is 21.375 g.

9. 4.55 $\mathbf{m}^2$

   Explanation: Multiply both dimensions by powers of 10 to convert to whole numbers (325 and 14), multiply, and then divide the result by 1000 (100×10):

$$3.25 \times 1.4 = \frac{325 \times 14}{1000} = \frac{4550}{1000} = 4.55 \text{ m}^2.$$

$$\begin{array}{r} {\scriptstyle 1\,2} \\ 325 \\ \times\ \ 14 \\ \hline 1300 \\ 3250 \\ \hline 4550 \end{array}$$

10. 35 **segments**

Explanation: Rewrite $15.75 \div 0.45$ as $1575 \div 45$ and divide:

$$\begin{array}{r} 35 \\ 45\overline{)157^{22}5} \end{array}$$

Thus, 35 segments can be cut from the spool.